

Machine Learning Based Prediction of COVID-19 Confirmed Cases in India

A Beginner-Friendly End-to-End Machine Learning Study

Abstract

This project demonstrates a simple end-to-end machine learning workflow for predicting cumulative COVID-19 confirmed cases in India. The supplied country-wise COVID-19 dataset is cleaned and filtered for India. The previous day's confirmed cases are used as the main feature. Three regression algorithms—Linear Regression, Decision Tree Regression and Random Forest Regression—are trained and compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared (R^2). The main purpose is educational: every step is kept simple enough for a beginner to understand and explain during a viva.

1. Introduction

Machine learning learns patterns from historical data and uses those patterns to make predictions. COVID-19 case counts are numerical, so regression is a suitable basic machine learning task. The supplied dataset contains day-to-day country-wise confirmed, death and recovered case information. This study focuses only on India so that the complete workflow remains easy to understand.

2. Problem Statement

The objective is to predict the current cumulative number of confirmed COVID-19 cases in India using the confirmed case count from the previous day. Since the target is a continuous numerical value, this is a regression problem.

3. Dataset

The project uses the supplied **full_grouped.csv** file. The original file contains 35,156 rows and 10 columns. After selecting India and preparing the previous-day feature, 187 observations are available for modelling. The data contains variables such as Date, Country/Region, Confirmed, Deaths, Recovered, Active and New cases.

4. Data Preprocessing

The Date column is converted from text into a date type. The India records are sorted chronologically. Duplicate rows are removed. Missing values in the main numerical columns are replaced with zero for this educational implementation. These steps make the data suitable for the modelling stage.

5. Feature Engineering

A new feature named Previous_Day_Confirmed is created with shift(1). For each row, it stores the confirmed case count from the previous day. The first row has no previous day and is therefore removed. This is a simple and understandable example of feature engineering.

6. Exploratory Data Analysis

Two basic graphs are produced. The first shows India's confirmed cases over time and helps identify the overall trend. The second is a scatter plot of previous-day confirmed cases against current confirmed cases and helps visualize their relationship.

7. Machine Learning Models

Three models are used. Linear Regression learns a straight-line relationship. Decision Tree Regression learns prediction rules by splitting the data into regions. Random Forest Regression combines many decision trees and can learn more complex relationships. The models are compared instead of assuming that one algorithm is always best.

8. Train-Test Split

The data is divided into 80% training and 20% testing data. Shuffling is disabled because the observations are chronological. This keeps earlier observations in training and later observations in testing, which is more sensible for this simple time-based problem.

9. Evaluation Metrics

MAE is the average absolute difference between actual and predicted values. RMSE is another error measure that penalizes large errors more strongly. R^2 indicates how much of the variation in the target is explained by the model. Lower MAE and RMSE are preferred, while higher R^2 is preferred.

10. Results

Model	MAE	RMSE	R^2
Linear Regression	3,048.88	3,782.19	0.9999
Decision Tree	442,304.63	541,164.94	-2.0121
Random Forest	451,907.29	549,041.26	-2.1005

In the generated run, **Linear Regression** produced the lowest RMSE and was selected as the best model. The exact metrics above were calculated from the supplied dataset using the beginner implementation.

11. Sample Prediction

After selecting the best model, a new sample is created with a previous-day confirmed count of 1,000,000. The model uses this input to produce a predicted current confirmed count. This demonstrates how a trained model can be used on unseen input.

12. Limitations

The model uses only one main feature and therefore is not a realistic epidemiological forecasting system. COVID-19 case counts are influenced by testing, reporting, interventions, behavior, population and many other variables. The project is intended as an academic demonstration of the machine learning workflow.

13. Conclusion

This project demonstrates the complete basic machine learning process: understanding the dataset, cleaning data, feature engineering, EDA, train-test splitting, training multiple models, evaluating them, selecting the best model and performing a sample prediction. The implementation is deliberately simple so that each line and concept can be explained clearly during a viva.

14. Future Scope

The project can later be improved by adding more lag features, additional countries, population-normalized values, testing data, cross-validation, regularization and hyperparameter tuning. These improvements should be added only after the basic workflow is understood.

References

1. Scikit-learn documentation: `train_test_split` and model evaluation.
2. Scikit-learn documentation: Linear Regression.
3. Kaggle COVID-19 Dataset / Corona Virus Report: description of the supplied country-wise daily COVID-19 data.